Archisha Ahuja

(Agricultural researcher focused on developing autonomous robotic systems for agricultural environments)

Github E-mail LinkedIn Project Website

Education

Year	Degree/Exam	Institute	GPA/%
2027	Agricultural and Food Engineering (B.Tech) (Micro-specialisation in Artificial Intelligence)	IIT Kharagpur	8.89/10

Research Interests

Robotics and Autonomous Systems, Computer Vision and Deep Learning for Precision Agriculture, Geospatial and Remote Sensing Analysis, Artificial intelligence Applications in Agri-Tech

Publications

- Sapkota, R., Bin Sabir, S. U., **Ahuja, A.**, & Karkee, M. **A wind-assisted active perception system for occluded green fruit detection and instance segmentation in commercial orchards** *Computers and Electronics in Agriculture* (Submitted)
- Sapkota, R., Bin Sabir, S. U., **Ahuja, A.**, & Karkee, M. **Semiautonomous system for robotic fruitlet thinning in commercial orchards** (In progress)

Conferences

- **Large-Scale Canopy Architecture Mapping in Apple Orchards-Poster, ASABE Annual International Meeting 2026.**

Research Internships

Greenfruit Thinning using UR5 Arm and UGV Integration [May'26-July'26]

Prof. Manoj Karkee, Cornell University

- **ROS-based pipeline** integrating a **UR5** arm (6dof) with a Warthog UGV for greenfruit thinning, using **Gazebo simulation**, **3D LiDAR navigation**, **grid-based visual servoing** (**sub-3 mm** accuracy) and **LiDAR–camera depth fusion** to handle sunlight.

Geospatial Classification of Apple Orchard Structures [Jun'25-Dec'25]

Prof. Manoj Karkee, Cornell University (Remote)

- **U-Net and Mask R-CNN** on NAIP satellite imagery to classify apple orchards into **Freestanding, Vertical, and V-Trellis** systems, achieving about **0.94** overall accuracy.

For more details on these research internships, visit the Project Website.

Projects

Vision-Based Precision Spraying Algorithm for Chilli Weed Detection [Mar'26-Sept'26]

Prof. Piyush Soni, IIT Kharagpur

- Real-time **weed detection and targeted-spraying pipeline** on an **NVIDIA Jetson Orin Nano Super** (DeepStream, up to **30 FPS**) that localizes weeds in 3D and drives an **ESP32**-controlled nozzle, pump and valves over UART.

Autonomous Weed Detection and Spraying Using GPS-Guided Drone [Jun'25-Dec'25]

Project Internship, TechnoS Instruments

- GPS-guided drone (**Raspberry Pi 4B, Pixhawk, YOLOv5-s**) that detects weeds in grass, geotags them and automates precision spraying, all controlled remotely over Wi-Fi.

Smart Agriculture Decision Support System [Sep'25-Oct'25]

Project, Smart India Hackathon

- Prototype AI decision-support system for farmers using **hyperspectral** and **sensor data**, with **CNN–LSTM** anomaly detection and forecasting, NDVI/NDWI-based irrigation and health maps, and an alert dashboard (wheat prototype).

For more details on these projects, visit the Project Website.

Skills and Expertise

Languages and Libraries	C, C++, Python, NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, OpenCV, SpaCy, QuantLib
Skills & Tools	Machine Learning, Supervised Learning, Deep Learning, NLP, EDA, Computer Vision, ROS2, Gazebo, Nav2, SLAM, Path Planning, 3D LiDAR, Depth Cameras, Visual Studio, Jupyter Notebook, Ubuntu, QGIS, Solidworks

Coursework

Agriculture: Soil Technology, Crop Production, Theory of Machine Elements, Farm Machinery, Land and Water Resources, Food Engineering, Tractor and Power Systems

AI & Robotics: Programming and Data Structures, Probability and Statistics, Linear Algebra, Image Processing, Linear Algebra for AI and ML, Artificial Intelligence: Foundations and Applications, AI/ML for Robot Autonomy, Mechanism and Robot Kinematics, CAD

Other: Basic Electronics, Partial Differential Equations, Advanced Calculus, Basic Thermodynamics, Fluid Mechanics

Positions of Responsibility

Digital Strategy & Outreach Lead 2026

Agricultural and Food Engineering Society (AFES), IIT Kharagpur

- Led digital communications, campus outreach, and team coordination to strengthen the visibility of the society among students and faculty.

Associate Quant Researcher 2024-2025

Quant Club, IIT Kharagpur

- Developed an LSTM-based portfolio optimization framework integrating stock price prediction, Monte Carlo simulations, and Efficient Frontier analysis, with hands-on exposure to derivatives, algorithmic trading, and backtesting.

Secretary 2021-22

Faraday Fraternity (Science Club), Jayshree Periwal High School

- Organized E-Science Fest and BOSA, coordinating 10 science-based events with international, national, and interschool outreach.

Extracurriculars

- Four consecutive Academic Excellence Awards and an All-Rounder Award, Jayshree Periwal High School.
- 1st position at IND Stage level in dance; completed a 5-year Kathak certification (Natraj Dance Academy); represented the school in interschool competitions.
- 1st place in Abhivyakti, the inter-college stage play competition at Thomso, IIT Roorkee; directed and acted in stage and street plays with Encore.